

Robust and interpretable laser-induced breakdown spectroscopy quantification of rare earth elements in polymetallic ores

Huihui Zhu, Jiawei Wang, Yubo Zhang, Xiaohui Su, Jian Wu, Xinyu Guo, Fuli Chen, Kaizhi Wang, Fangfang Wang, Tao Lü



PII: S0026-265X(26)02666-4

DOI: <https://doi.org/10.1016/j.microc.2026.119461>

Reference: MICROC 119461

To appear in: *Microchemical Journal*

Received date: 19 April 2026

Revised date: 11 August 2026

Accepted date: 19 August 2026

Please cite this article as: H. Zhu, J. Wang, Y. Zhang, et al., Robust and interpretable laser-induced breakdown spectroscopy quantification of rare earth elements in polymetallic ores, *Microchemical Journal* (2026), <https://doi.org/10.1016/j.microc.2026.119461>

This is a PDF of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability. This version will undergo additional copyediting, typesetting and review before it is published in its final form. As such, this version is no longer the Accepted Manuscript, but it is not yet the definitive Version of Record; we are providing this early version to give early visibility of the article. Please note that Elsevier's sharing policy for the Published Journal Article applies to this version, see: <https://www.elsevier.com/about/policies-and-standards/sharing#4-published-journal-article>. Please also note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

Robust and interpretable laser-induced breakdown spectroscopy quantification of rare earth elements in polymetallic ores

Huihui Zhu,^{1,2} Jiawei Wang,^{1,2} Yubo Zhang,^{1,2} Xiaohui Su,³ Jian Wu,⁴ Xinyu Guo,⁴ Fuli Chen,^{1,2} Kaizhi Wang,³ Fangfang

Wang,⁵ and Tao Lü^{1,2, a)}

¹ School of Artificial Intelligence and Automation, China University of Geosciences, Wuhan 430074, China

² Hubei Key Laboratory of Advanced Control and Intelligent Automation for Complex Systems, Wuhan 430074, China

³ School of Future Technology, China University of Geosciences, Wuhan 430074, China

⁴ State Key Laboratory of Electrical Insulation and Power Equipment, Xi'an Jiaotong University, Xi'an, 710049, China

⁵ School of Mathematics and Physics, China University of Geosciences, Wuhan 430074, China

^{a)} E-mail: lvtaohn@126.com

ABSTRACT: Quantitative analysis of rare earth elements (REEs) in ores by laser induced breakdown spectroscopy (LIBS) remains challenging, as differences in matrix composition can modify laser ablation and plasma excitation behavior and thereby compromise the stability of spectral-concentration relationships. A fusion model was developed for quantification of REEs by combining the local feature extraction capability of convolutional neural networks (CNN) with the long-range correlation ability of a Transformer, with the aim of improving model adaptability to matrix variation. Across the 16 single-matrix tasks derived from four matrices, the fusion model yielded the lowest RMSE in 13 tasks and showed narrower error fluctuations across matrices. The fusion model also achieved the best predictive performance ($R^2 = 0.9963$, RMSE = 0.418 wt.%) for Yb in mixed-matrix conditions. Gradient-weighted class activation mapping (Grad-CAM), wavelength-contribution analysis, spectral zone SHapley Additive exPlanations (SHAP) analysis, and t-distributed stochastic neighbor embedding (t-SNE) were employed to examine the spectroscopic evidence and feature representations associated with the model predictions, including whether the model used analyte related spectral regions while reducing clustering dominated by matrix differences. The results suggest that the combination of local and global spectral modeling provides a more robust and accurate strategy for LIBS quantification of REEs in polymetallic ores under matrix variation.

Keywords: laser-induced breakdown spectroscopy, rare earth elements, CNN-Transformer, attribution analysis, feature-space representation

1. Introduction

Rare earth elements (REEs) are of broad technological importance in fields such as permanent magnets[1,2], nuclear safety[3], and lasers and optical communications[4,5]. In polymetallic ores and related host-rock systems, REEs are also important geochemical indicators and economically relevant components for resource evaluation, ore sorting, and beneficiation design[6,7]. Reliable quantification of REEs in complex ores is therefore important not only for geochemical interpretation, but also for practical decision-making in mineral exploration and utilization.

Laser-induced breakdown spectroscopy (LIBS) has attracted sustained interest for geological and nuclear-related applications because it enables rapid, multi-element, and in situ analysis[8–10]. In LIBS, a pulsed laser generates a plasma on the sample surface, and the emitted radiation is collected for elemental identification and concentration prediction. Compared with conventional chemical methods, LIBS is particularly attractive for solid samples because it combines minimal sample preparation with high analytical throughput and field applicability[11].

Despite these advantages, quantitative LIBS analysis of REEs in polymetallic ores remains difficult. One source of difficulty is the intrinsic spectroscopic complexity of REEs. Their partially filled $4f$ shells give rise to numerous possible radiative transitions, producing dense and strongly overlapping emission lines[12,13]. In complex ores, this intrinsic complexity is further compounded by matrix-dependent effects[14]. Common matrix components such as Fe, Si, Al, and Ca not only introduce additional emission lines and continuum background, but also alter ablation efficiency, plasma temperature, and electron density, thereby changing the emission response of the target analytes[14–16]. As a result, the relationship between spectral intensity and elemental concentration often becomes unstable from one ore matrix to another, and calibration strategies based on a few selected lines are easily degraded[17,18].

To alleviate these limitations, different calibration and modeling strategies have been explored. Traditional chemometric methods, including partial least squares regression (PLSR), support vector regression (SVR), and random forest, can exploit multivariate spectral information and generally improve upon univariate line-based calibration[12,17,19]. More recently, deep-learning models have been introduced into LIBS quantification to learn nonlinear spectral-concentration relationships from full spectrum inputs[20,21]. Convolutional neural network (CNN), attention, and Transformer based models have shown improved prediction performance by using richer spectral features than isolated analytical lines for LIBS quantification of REEs in polymetallic ores[9,22]. However, most recent studies have primarily focused on improving prediction accuracy within a given dataset or matrix setting, without investigating a more practical question in complex ore analysis, whether a model can maintain reliable quantitative performance when matrix composition changes. In polymetallic ores, matrix variations can affect laser ablation and plasma excitation behavior, thereby altering the background level and local peak shape of the spectrum[14–16,23,24].

This problem is difficult to address with a single modeling mechanism. In LIBS spectra of REEs, useful analyte information appears at different spectral scales. At the local scale, dense lanthanide lines

form adjacent peaks, peak clusters, and partially overlapped structures, for which CNNs are well suited because they can encode peak-shape and neighborhood features along the wavelength axis. At the global scale, the concentration response of one REE is often distributed among multiple atomic and ionic emission lines located in different wavelength regions, for which Transformer based models are advantageous because they can capture long-range dependencies across distant spectral bands[20,21,25,26]. This motivates a local-global fusion model in which CNNs first extract spectroscopically structured local tokens and the Transformer subsequently models correlations among spectrally separated regions. At the same time, improved prediction accuracy alone does not prove that a full spectrum model has learned analyte-relevant spectral information. Under multi-matrix conditions, part of the apparent prediction gain may arise from matrix specific background variation, continuum changes, or sample-group differences that are correlated with concentration[27–30]. Therefore, model interpretation is necessary to examine both the wavelength regions supporting the prediction and the feature-space representation learned by the model. For LIBS quantification of REEs, this is especially important because the model should rely on spectroscopically meaningful emission regions of the target elements rather than on spectral features dominated by matrix differences.

To address these issues, this work adapts a CNN-Transformer framework to the specific analytical problems encountered in LIBS quantification of REEs in polymetallic ores. The contribution lies in organizing local spectral encoding and global correlation modeling according to the spectral characteristics of REEs. The analytical objective of this study is not limited to improving prediction accuracy within a single dataset. Gradient-weighted class activation mapping (Grad-CAM), wavelength contribution analysis, spectral zone SHapley Additive exPlanations (SHAP) analysis, and t-distributed stochastic neighbor embedding (t-SNE) were used to examine the model from two complementary perspectives, namely wavelength level evidence and feature space representation, and to assess whether the model uses spectroscopically relevant analyte information while reducing feature separation dominated by matrix differences.

2. Materials and methods

2.1 Samples

The study selected three ore materials with different matrices for analysis, GBW07370 (Sn-Pb-Cu ore, ~69% TFe_2O_3), GBW07369 (W-Sn-Bi ore, ~48% SiO_2), and uranium ore (U, ~70% SiO_2). The uranium ore samples were further divided into two subsets (U-1 and U-3) according to differences in U content. **Table 1** lists the major constituents of the four matrices. To obtain samples spanning different concentration levels of the target analytes, rare earth oxides (Sm_2O_3 , Gd_2O_3 , Yb_2O_3 , Nd_2O_3 , Dy_2O_3 , Pr_2O_3 and Er_2O_3) were spiked into the respective ore matrices at predetermined ratios to adjust the corresponding rare earth element contents. The samples were thoroughly ground in an agate mortar, followed by the addition of 5-6 drops of 10% poly (vinyl alcohol) solution as a binder, and then further ground for 20-30 min until completely dry and homogeneous. The resulting powders were pressed into pellets (15 mm in diameter) for laser-induced breakdown spectroscopy (LIBS) analysis, as shown in

Fig.S1 (a). In total, 40 samples were prepared, and the rare earth element concentrations, expressed as elemental mass fractions in wt.%, together with the line-based limits of detection (LoDs) for each combination of element and matrix, are summarized in **Table S1 and Table S2**. The concentration ranges of each REE in each matrix are summarized in **Table S6**. Two experimental schemes were considered: a single-matrix scheme and a mixed-matrix scheme. In the single-matrix, independent datasets were constructed for GBW07369, GBW07370, and each of the two uranium ores (10 samples per subset), and model training and testing were conducted within each matrix. In the mixed-matrix, all 40 samples from the four matrix subsets were pooled for model development and validation. This evaluation was performed for Yb because it was the only REE present in all four matrices.

Table 1. Major components of the four base matrices. All values are reported in wt.% unless otherwise specified. U concentrations marked with an asterisk are reported in ppm.

Primary component	GBW07370	GBW07369	Uranium ore 1 (U-1)	Uranium ore 3 (U-3)
SiO ₂	0.98	48.2	73.54	66.19
Al ₂ O ₃	0.45	8.43	11.04	15.09
Fe ₂ O ₃	67.80	8.44	3.00	2.98
MgO	0.35	1.06	1.58	1.46
CaO	1.69	9.18	2.13	3.34
Na ₂ O	0.02	0.96	1.78	2.82
K ₂ O	0.02	1.85	4.09	5.16
MnO	0.47	1.20	0.06	0.06
W	0.03	7.96	-	-
Bi	0.03	1.78	-	-
Sn	9.56	0.06	-	-
U*	29.8*	68.5*	92.8*	860*

2.2 LIBS measurement

The LIBS experiments employed a nanosecond-pulsed laser (DAWA 200, Beamtech Optronics, Beijing) operating at 1064 nm with an 8 ns pulse width and 30 mJ pulse energy, as shown in **Fig. S1(b)**. The beam was focused by a focusing objective ($f = 30$ mm) to a point 200 μ m below the sample surface to avoid premature air breakdown. During plasma evolution, emission was collected by a lens, coupled into an optical fiber and delivered to an echelle spectrometer (ARYELLE, LTB) equipped with an intensified charge coupled device (ICCD, DH334T, Andor). The echelle spectrometer covered the wavelength range of 265-717 nm, and the full-spectra within this range were used for subsequent analysis. Timing between the laser and spectrometer was controlled by a digital delay generator (DG645, Stanford Research Systems). By comparing time-resolved acquisition data obtained under different gate delays and gate widths, and using the signal-to-noise ratio as the optimization criterion, we determined a gate delay of 3 μ s and a gate width of 5 μ s as the optimal parameter combination. When the gain was set to 1500, it provided sufficient spectral intensity while avoiding spectral saturation at high concentrations. To mitigate elemental heterogeneity and shot-to-shot fluctuation, 20 consecutive laser

shots were accumulated at one sampling position to form a single spectrum. This acquisition was repeated at 20 different positions on each pellet, yielding 20 replicate spectra per sample for subsequent analysis. The recorded full spectrum intensities were directly used as inputs for all models to enable a consistent end-to-end comparison.

2.3 Methodology

2.3.1 Convolution-based local feature extraction

LIBS spectra are inherently one-dimensional sequences arranged along the wavelength axis, and adjacent wavelengths usually exhibit strong local correlations, particularly in regions with severe line overlap and dense peak clusters. Based on this characteristic, a one-dimensional CNN was used to extract local spectral features [31]. The input spectrum was organized as a single-channel tensor $[B, 1, L]$ in wavelength order, so that the neighborhood structure of spectral peaks along the wavelength axis was preserved.

In the feature extraction module, a three-stage CNN was used to hierarchically encode local spectral patterns. The first stage used a relatively broad receptive field to capture local peak shapes and neighborhood background fluctuations, whereas the following stages progressively reduced the convolutional scale and increased the channel dimension to learn peak clusters, overlapped line structures, and weaker matrix-interference patterns. Down-sampling operations compressed the spectral sequence and enlarged the effective receptive field before regression or subsequent Transformer modeling.

After feature extraction, the final feature output by the CNN is denoted as

$$H = \Phi(x) \in \mathbb{R}^{B \times C \times L'} \quad (1)$$

where B is the batch size, $C = 128$ is the number of channels in the last layer, and L' is the spectral axis length after three stages down sampling. The feature is then flattened and fed into parallel regression heads for different elements to complete multi-task quantitative prediction:

$$\hat{y}^{(t)} = w_t^T \text{vec}(H) + b_t, \quad t = 1, \dots, T, \quad (2)$$

where T is the number of target elements, and w_t and b_t are the parameters of the linear regression head corresponding to the t element. During training, the overall loss was defined as the weighted sum of the mean squared error (MSE) losses:

$$\mathcal{L} = \sum_{t=1}^T \alpha_t \frac{1}{B} \sum_{n=1}^B (\hat{y}_n^{(t)} - y_n^{(t)})^2 \quad (3)$$

where $\mathcal{L}$ is the total loss and α_t is the weight assigned to task t . Before model training, each target concentration was normalized using the training set statistics, and equal task weights were used for the normalized targets. The reported RMSE values were calculated after inverse transformation to wt.%.

2.3.2 Transformer-based global feature extraction

Unlike CNNs, which hierarchically aggregate local features, Transformers are better suited to modeling correlations between spectrally distant wavelength regions. In LIBS spectra, such long-range

dependencies arise from coordinated variations among different emission lines of the same element, as well as from global spectral-shape changes induced by matrix effects. On this basis, a Transformer model was further introduced to extract global features.

Specifically, the input one-dimensional spectrum was first partitioned into fixed-length patches along the wavelength axis. Each patch was then projected by a linear layer into a d_{model} dimensional token, followed by LayerNorm to improve training stability. This design preserves local band information while reducing the sequence length, thereby avoiding the high computational cost of applying attention directly to long spectral sequences.

Because self-attention is inherently insensitive to input order, sinusoidal absolute positional encoding was introduced to retain positional information along the wavelength axis[32]. Let the sequence length input to the Transformer encoder be S , and for position $p \in \{0, \dots, S-1\}$, the positional encoding PE is defined as:

$$PE_{(p,2j)} = \sin\left(\frac{p}{10000^{\frac{2j}{d_{model}}}}\right) \quad (4)$$

$$PE_{(p,2j+1)} = \cos\left(\frac{p}{10000^{\frac{2j}{d_{model}}}}\right) \quad (5)$$

where $j = 0, \dots, \frac{d_{model}}{2} - 1$. The final sequence fed into the Transformer encoder is $H_0 = Dropout(Z_0 + PE)$. This construction provides each token position with a unique and distinguishable representation, enabling the model to differentiate peak-interference structures with similar morphology but located at different wavelengths, and to leverage positional information in subsequent attention computations to model cross-band correlations. The feature-extraction layer adopts a multi-head self-attention mechanism to capture the coupling relationships among emission lines of the same element across the full-spectral range. For the l layer, the input sequence H_{l-1} is first projected into the query (Q), key (K), and value (V) matrices:

$$Q = H_{l-1}W^Q, \quad K = H_{l-1}W^K, \quad V = H_{l-1}W^V \quad (6)$$

where W^Q , W^K , W^V are three trainable parameter matrices.

Then, the global attention weight matrix is computed as[32]:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}} + M\right) \quad (7)$$

where d_k is the feature dimension of the key. M is a masking term derived from the key padding mask m . The multi-head attention output is subsequently processed by linear projection, residual connection, and layer normalization, and then further transformed by a feed-forward network (FFN). For the standalone Transformer model, the encoded spectral sequence was aggregated according to the implementation used in the corresponding baseline.

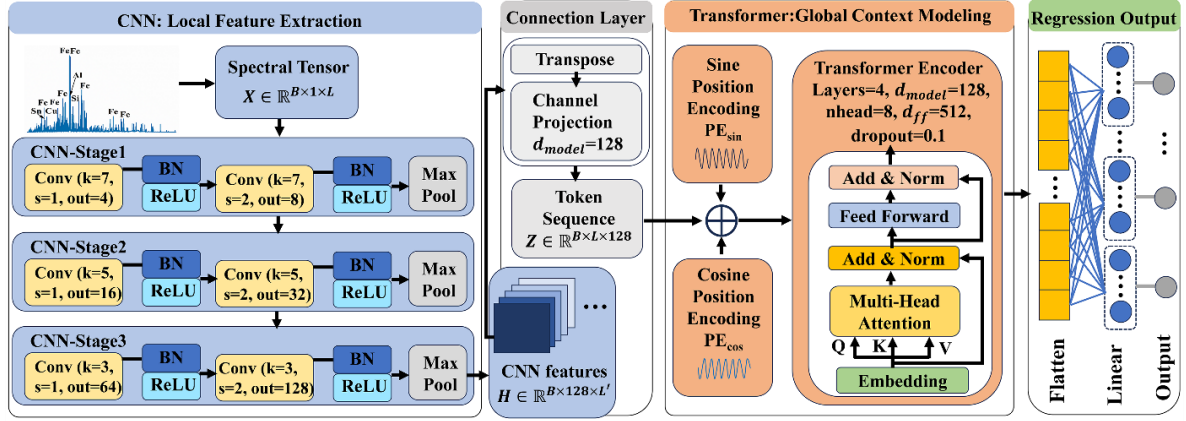


Fig. 1. Flowchart of CNN-Transformer fusion model.

2.3.3 CNN-Transformer fusion model

The proposed CNN-Transformer fusion model was designed to combine local feature extraction by CNNs with global correlation modeling by the Transformer in **Fig. 1**. This hybrid architecture is particularly suitable for LIBS spectra of REEs, where severe line overlap requires convolution to first capture local peak patterns and clustered spectral features, while the relationships between distant wavelength regions need to be modeled by the attention mechanism. For fair comparison, the standalone CNN and Transformer models were designed to match the two branches of the fusion framework and were trained separately with their own regression heads.

In the fusion architecture, the CNN branch first compresses the high dimensional full spectrum into a shorter feature sequence. This sequence is then transposed and projected to the Transformer embedding dimension, after which self-attention is used to model correlations among spectrally separated regions. The Transformer output sequence is subsequently flattened and passed to independent regression heads for the target elements. Detailed architectural and training parameters are provided in **Table S5**.

2.3.4 Group-level leave-one-out validation

To maximize the use of training data under limited sample availability, while ensuring that replicate spectra from the same physical sample were not simultaneously included in the training and test sets, a group-level leave-one-out cross-validation (LOOCV) was employed. The experimental dataset covers four different matrices (GBW07369, GBW07370, U-1, U-3). Each matrix contains 10 independent physical samples with different concentration gradients, giving 40 samples in total. For each physical sample, 20 replicate spectra were collected, yielding 800 spectra overall. During validation, the dataset was divided at the level of physical sample. In each fold, all 20 replicate spectra from one physical sample were held out for testing, while the remaining physical samples were used for model training. The model was reinitialized before each fold. For each matrix-specific evaluation, one physical sample (20 replicate spectra) served as the independent test set and the remaining nine samples (180 spectra) served as the training set. This procedure was repeated over the 10 samples in each matrix. In the mixed-matrix condition, one physical sample (20 spectra) was used for testing and the remaining 39 samples

(780 spectra) were used for training, yielding 40 folds in total.

2.3.5 Model evaluation

To evaluate predictive accuracy and robustness of different models under different matrix conditions, the root mean square error (RMSE) and the coefficient of determination (R^2) were used as the primary evaluation metrics. RMSE characterizes the average deviation between predicted and true values, while R^2 reflects the linear consistency between predicted and true values (values closer to 1 indicate a better fit). Because the reference concentrations of the target REEs were expressed in wt.%, all RMSE values reported in this work are also expressed in wt.%. Let the test set of the m matrix ($m = 1, \dots, M, M = 4$) contain n_m prediction observations, with true values $y_{m,i}$ and predictions $\hat{y}_{m,i}$ ($i=1, \dots, n_m$)

$$RMSE_m = \sqrt{\frac{1}{n_m} \sum_{i=1}^{n_m} (y_{m,i} - \hat{y}_{m,i})^2} \quad (8)$$

$$R_m^2 = 1 - \frac{\sum_{i=1}^{n_m} (y_{m,i} - \hat{y}_{m,i})^2}{\sum_{i=1}^{n_m} (y_{m,i} - \bar{y}_m)^2} \quad (9)$$

where $\bar{y}_m = \frac{1}{n_m} \sum_{i=1}^{n_m} y_{m,i}$ is the mean of the true values in the test set of the m matrix. The reported R^2 and RMSE values were calculated by pooling the replicate spectrum predictions from the held out physical samples across all cross-validation folds. rRMSE was calculated as RMSE divided by the mean reference concentration of the test observations and expressed as a percentage.

Furthermore, to quantify performance fluctuations across different matrices (sensitivity to matrix effects), the $RMSE_m$ values obtained for the four matrices are aggregated, and the standard deviation (SD) of RMSE across matrices and the corresponding coefficient of variation (CV) are computed:

$$SD = \sqrt{\frac{1}{M-1} \sum_{m=1}^M (RMSE_m - \overline{RMSE})^2} \quad (10)$$

$$CV = \frac{SD}{\overline{RMSE}} \quad (11)$$

Here, SD represents the absolute magnitude of RMSE fluctuations across matrices. CV normalizes this fluctuation by the mean error and therefore reflects the relative degree of variation. When the average error level is comparable, a smaller CV indicates more consistent performance across matrices.

To evaluate the proposed model, two conventional regression models, PLSR and SVR, were included for comparison. All models were trained using the same group-wise data partitioning strategy, and their hyperparameters were optimized independently using only the training data. The deep learning models were trained with AdamW using a learning rate of 1×10^{-5} and a batch size of 16 to improve training stability. The networks were implemented in PyTorch 2.7.0 with Python 3.12.7 and trained on

an NVIDIA GPU under CUDA 12.8.

3. Results and discussion

3.1 Limitations of line-based calibration in REE analysis under different matrices

3.1.1 Matrix dependence of univariate calibration

To evaluate the accuracy and robustness of line-based quantification under different matrix conditions, univariate calibrations were first performed for Sm, Gd, Yb, and Er in two representative matrices, GBW07369 and GBW07370, and the results are summarized in **Table S3**. To further facilitate comparison between matrices, the mean RMSE across the selected lines and the best-line RMSE for each element in each matrix were additionally summarized in **Table S4**. Within each individual matrix, several characteristic lines still showed usable linear relationships. However, for the same element, quantitative performance differed markedly between matrices. For Sm, Yb, and Er, both the mean RMSE and the best-line RMSE were higher in GBW07370 than in GBW07369, whereas Gd showed the opposite trend. This indicates that prediction error remained matrix dependent even after selecting the best-performing line within each matrix, suggesting that line-based calibration may have limited transferability across compositionally different ore matrices.

3.1.2 Spectroscopic origins of this limitation

This limitation can be understood from two aspects. First, REEs exhibit intrinsically complex spectral structures. As shown in **Fig. S2**, Pr, Nd, Sm, Gd, Dy, Er, and Yb, together with their singly ionized species, have numerous possible radiative transitions arising from their partially filled 4f shells. As a result, LIBS spectra of REEs usually contain dense and closely spaced emission lines, and analyte-related information is often distributed over multiple coupled peaks and peak clusters rather than concentrated in a few isolated lines.

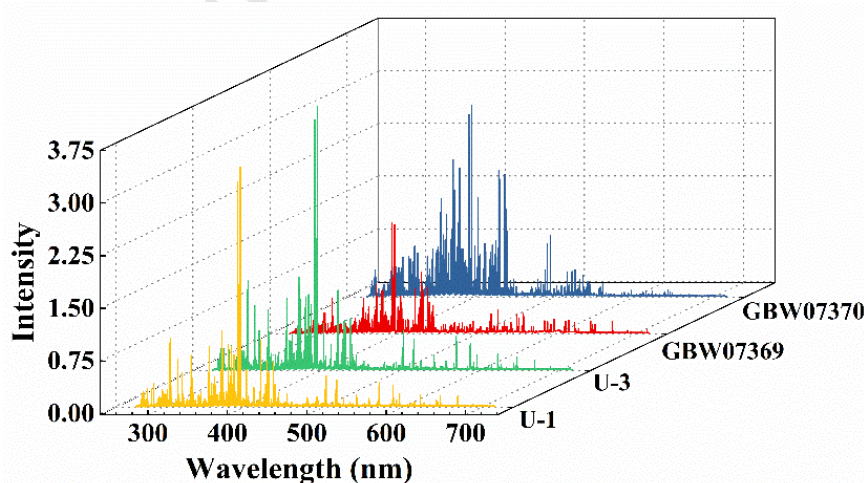


Fig. 2. Representative LIBS emission spectra of samples from different matrices (U-1, U-3, GBW07369, and GBW07370)

In polymetallic ore samples, these factors increase the difficulty of line assignment and weaken the stability of line-based spectral-concentration relationships. Under such conditions, calibration strategies that rely on only a limited number of characteristic lines are unlikely to maintain stable

performance in complex ore systems.

Second, this intrinsic complexity is further compounded by matrix-dependent spectral variation. As shown in **Fig. 2**, representative raw LIBS spectra from the four matrices exhibited clear differences in background level, emission density, and local peak distribution. The Fe-rich matrix GBW07370 showed a higher continuum background and denser emission features, particularly in the 350–450 nm region. In U-1 and U-3, the more prominent emission features were also mainly concentrated in this region, whereas the GBW-series samples retained more pronounced peak clusters at longer wavelengths. These differences indicate that matrix composition affects not only the background level but also the local spectral environment and peak shape.

3.2 Single-matrix quantitative performance of competing models

3.2.1 Analysis of model prediction performance

Two conventional chemometric models (PLSR and SVR) and three deep learning models (CNN, Transformer, and CNN-Transformer) were compared, with emphasis on predictive accuracy and error distribution. **Table 2** summarizes the prediction results of the five models for each single-matrix dataset. Compared with the univariate calibrations shown in **Table S3**, both PLSR and SVR achieved higher R^2 values and lower RMSE values for most tasks, indicating that the incorporation of full spectrum information enables more effective characterization of concentration-related spectral features. Among the two conventional baselines, SVR outperformed PLSR in most cases, suggesting that nonlinear regression is better able to accommodate matrix-related spectral variation.

Table 2 Comparison of quantitative prediction performance for REEs across different matrices (Matrix-1 to Matrix-4). The R^2 and RMSE are reported for PLSR, SVR, CNN, Transformer, and CNN-Transformer, and all RMSE values are in wt.%.

Matrices	Elements	PLSR		SVR		CNN		Transformer		CNN-Transformer	
		R^2	RMSE	R^2	RMSE	R^2	RMSE	R^2	RMSE	R^2	RMSE
Matrix-1	Sm	0.942	1.457	0.944	1.431	0.967	1.119	0.816	2.634	0.993	0.506
	Gd	0.857	1.923	0.817	2.173	0.884	1.727	0.726	2.654	0.932	1.320
	Er	0.847	1.782	0.860	1.706	0.960	0.916	0.912	1.354	0.994	0.345
	Yb	0.908	1.625	0.931	1.404	0.992	0.468	0.908	1.627	0.994	0.421
	Pr	0.924	1.541	0.935	1.423	0.991	0.541	0.976	0.861	0.981	0.778
Matrix-2	Sm	0.918	0.786	0.926	0.748	0.984	0.349	0.935	0.704	0.986	0.322
	Gd	0.899	1.316	0.931	1.093	0.948	0.945	0.962	0.810	0.979	0.607
	Er	0.840	0.811	0.815	0.872	0.960	0.407	0.940	0.496	0.983	0.260
	Yb	0.789	0.941	0.735	1.054	0.985	0.248	0.901	0.644	0.952	0.451
	Dy	0.871	1.324	0.922	1.031	0.988	0.404	0.877	1.295	0.995	0.252
Matrix-3	Nd	0.926	2.157	0.871	2.844	0.981	1.085	0.985	0.953	0.988	0.879
	Dy	0.875	2.743	0.927	2.095	0.974	1.254	0.937	1.959	0.984	0.972
	Yb	0.897	2.331	0.918	2.084	0.965	1.359	0.924	2.006	0.990	0.676
Matrix-4	Nd	0.923	2.347	0.882	2.894	0.978	1.240	0.936	2.142	0.983	1.101
	Dy	0.861	3.005	0.941	1.955	0.987	0.909	0.888	2.696	0.940	1.982
	Yb	0.895	2.197	0.908	2.061	0.813	2.939	0.921	1.913	0.956	1.521

The deep learning models further improved prediction performance relative to the conventional

chemometric baselines. Across the 16 single matrix tasks, the CNN-Transformer achieved the lowest RMSE in 13 cases and showed the best overall performance. The three exceptions were Pr in Matrix 1, Yb in Matrix 2, and Dy in Matrix 4, for which CNN gave lower RMSE values. Because these cases involved different elements and matrices, they did not indicate a consistent element specific or matrix specific pattern. The relative value of local and global information can vary across quantitative tasks as the distribution of analyte related features, local interference, and background structure changes among element and matrix combinations. These task level differences did not alter the overall trend, with the CNN-Transformer achieving the lowest RMSE in most tasks.

3.2.2 Analysis of model robustness

For robustness evaluation, the boxplots in **Fig. S3** compare the RMSE distributions across matrices for the three deep learning models. Transformer shows larger spreads and more extreme tails for several elements, most notably Yb, Dy, and Nd, indicating less stable performance as the matrix changes. CNN, although accurate in some cases, produced higher error levels and more obvious dispersion in the more difficult matrices, particularly Matrix-3 and Matrix-4, suggesting stronger sensitivity to matrix-dependent background and interference. By contrast, the RMSE distributions of the CNN-Transformer model were generally more compact, with shorter interquartile ranges and fewer extreme values, indicating more consistent prediction errors under matrix variation.

To examine the incremental effect of global correlation modeling on a common target, the Yb RMSE values across the four matrices were further summarized for CNN and CNN-Transformer using the mean RMSE, SD, and CV. Using the Yb RMSE values in Table 2 and the sample SD definition in Eq. (10), CNN gave a mean RMSE of 1.254 wt.%, SD of 1.222 wt.%, CV of 0.975 and CNN-Transformer gave a mean RMSE of 0.767 wt.%, SD of 0.515 wt.%, CV of 0.672. These results indicate that the fusion model reduced both the average error and the absolute RMSE dispersion across matrices relative to CNN.

3.3 Mixed-matrix evaluation using Yb as the common analyte

The single-matrix results showed that CNN-Transformer generally achieved lower prediction error and smaller cross-matrix fluctuations than the compared models. A more demanding evaluation is whether this advantage can be maintained when samples with different matrix compositions are analyzed within a mixed-matrix dataset. Under this setting, differences in ablation behavior, plasma state, and local spectral environment further complicate the stability of concentration-spectral relationships across samples. Yb was selected as the representative target because it was the only analyte shared by all four matrices. The predictive performance of CNN, Transformer, and CNN-Transformer under the mixed-matrix setting is compared in **Fig. 3**. CNN showed the largest scatter and the weakest agreement with the diagonal trend, particularly in the medium- and high-concentration ranges ($R^2 = 0.8930$, RMSE = 2.259 wt.%, rRMSE = 24.67%). Transformer substantially improved the prediction distribution, with points clustered much more closely around the diagonal ($R^2 = 0.9948$, RMSE = 0.497 wt.%, rRMSE = 5.44%). CNN-Transformer yielded the closest agreement between predicted and

measured values and achieved the best overall performance ($R^2 = 0.9963$, RMSE = 0.418 wt.%, rRMSE = 4.58%). These results suggest that combining local spectral shape extraction with global correlation modeling is beneficial for Yb quantification within the mixed-matrix setting.

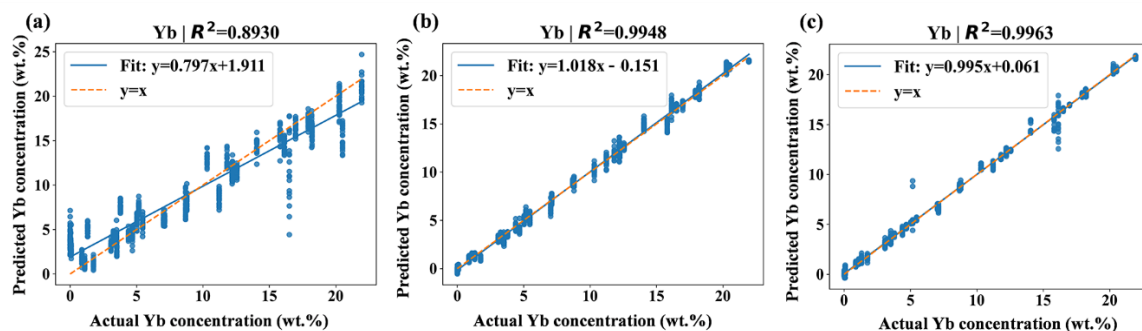


Fig. 3. Predicted versus measured Yb concentrations under the mixed-matrix for (a) CNN, (b) Transformer, and (c) CNN-Transformer.

While **Fig. 3** reflects the overall agreement between predicted and measured values, **Fig. S4** further illustrates within matrix repeatability and the separation between individual samples. Because each sample contained 20 replicate spectra, the predictions within each matrix formed several approximately parallel distribution bands corresponding to different sample levels. The thickness of each band reflects the dispersion among replicate measurements, and the overlap between adjacent bands reflects sample separability in the prediction space. CNN produced relatively broad bands with frequent overlap between neighboring samples, indicating larger repeatability fluctuations and weaker discrimination among samples. Transformer showed a more concentrated distribution than CNN, although noticeable dispersion remained in some matrices. By contrast, CNN-Transformer exhibited narrower bands and less overlap across all four matrices, indicating improved agreement, repeatability, and sample discrimination under the mixed-matrix condition. Because Yb was the only analyte shared by all four matrices, the mixed-matrix result is limited to Yb.

3.4. Feature-space analysis

To determine whether the differences observed in the mixed-matrix Yb evaluation were also reflected in the learned representations, t-SNE was applied to the raw spectra and to the intermediate features before the regression head [33], as shown in **Fig. 4**. In the raw spectral space, samples were mainly separated by matrix type, indicating that the original feature space was strongly influenced by matrix-related variation. After feature extraction, this matrix-driven separation became weaker. CNN features still retained visible matrix clustering, whereas Transformer and CNN-Transformer features showed substantially stronger cross-matrix overlap, with the fusion model exhibiting the weakest matrix separation.

To complement the t-SNE visualization, matrix label separability was further quantified on the corresponding high dimensional feature vectors using the silhouette score and a linear-probe classification accuracy. The matrix label silhouette scores for RAW, CNN, Transformer, and CNN-Transformer were 0.248, 0.226, 0.0532, and 0.036, respectively, and the corresponding linear-probe

matrix-classification accuracies were 0.944, 0.903, 0.825, and 0.808. These quantitative results are consistent with the visual trend in **Fig. 4** and indicate that CNN-Transformer yielded the lowest matrix label separability among the compared representations, with part of the matrix context still retained in the learned features. The reduced matrix-label separability indicates weaker matrix dominated clustering in the learned representation. The silhouette score measures the overall clustering tendency according to matrix labels, whereas the linear probe evaluates whether matrix labels remain linearly recoverable from the high-dimensional features. Linear-probe accuracy alone does not define an acceptable level of residual matrix information and should be interpreted together with the quantitative prediction results. In the pooled mixed-matrix evaluation, the CNN-Transformer showed the lowest matrix-label separability while maintaining the best Yb prediction performance among the compared models.

Overall, CNN-Transformer provided the best predictive performance for Yb under the mixed-matrix condition. Compared with CNN and Transformer, the fusion model achieved closer agreement between predicted and measured values and lower prediction dispersion across samples from different matrices. Together with the attribution analysis, the feature space results indicate that CNN-Transformer reduced matrix dominated separation while preserving spectral information relevant to concentration prediction.

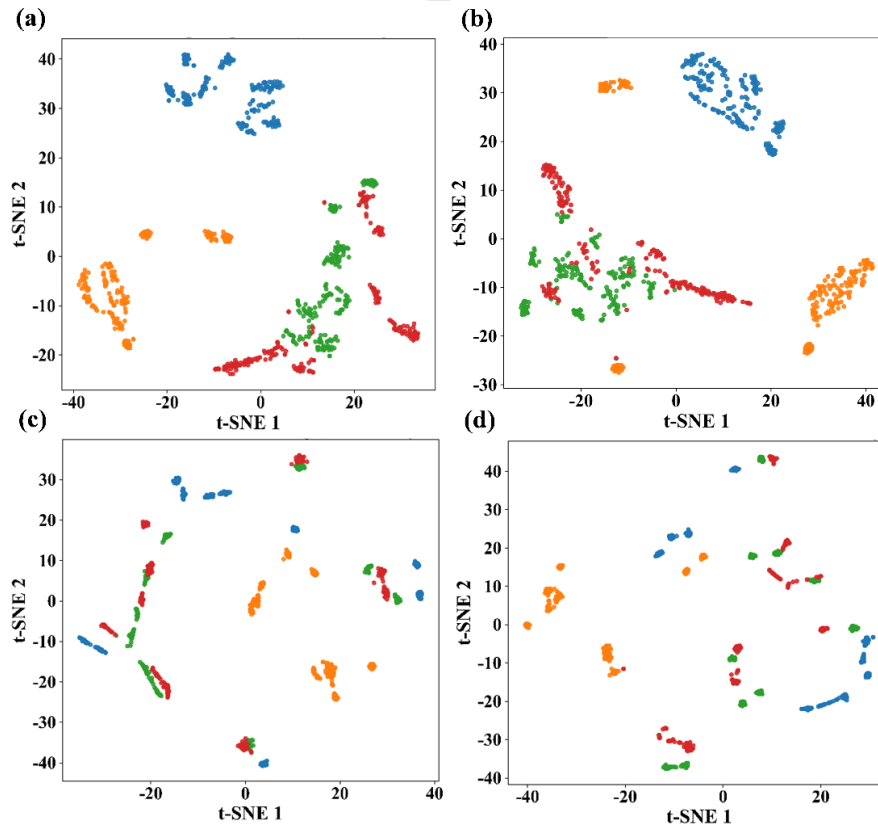


Fig. 4. t-SNE visualization of different feature representations to examine matrix-dependent clustering under multi-matrix conditions. (a) Raw full spectrum inputs, intermediate features extracted by the (b) CNN, (c) Transformer and (d) CNN-Transformer. Colors denote different matrix types. Blue: Matrix-1, Orange: Matrix-2, Green: Matrix-3, Red: Matrix-4.

3.5 Attribution analysis of Yb related spectral evidence across matrices

To better understand why CNN-Transformer showed smaller error fluctuations in the single-matrix evaluation and stronger prediction performance in the mixed-matrix condition, the spectral evidence underlying its predictions was further analyzed. Because the Transformer branch operates on convolutional feature sequences, attribution analysis was first performed on the shared convolutional backbone to localize the main informative wavelength regions retained at the front end of the fusion model. Positional contribution profiles and selected wavelength windows were then compared to examine how this evidence varied across matrices. Since Yb was the only analyte shared by all four matrices and also served as the representative target in the mixed-matrix evaluation, the cross-matrix attribution analysis focused on Yb.

Grad-CAM was first applied to localize wavelength regions relevant to the regression output. To obtain feature responses closer to the final regression decision, the last convolutional layer in the shared feature extractor (Conv index = 17) was selected as the target layer. Representative results for different elements in Matrix-1 are shown in **Fig. S5**. High-response regions were concentrated mainly between 300 and 500 nm, with particularly strong responses in the 350-450 nm range. Comparison with the spectra of pure rare-earth oxides showed that these high-response regions largely overlapped with spectral segments containing dense lines and prominent peak clusters. This observation suggests that the front-end feature extractor assigned greater importance to wavelength regions associated with elemental emission and stronger discriminative value. A high attribution weight at a given wavelength does not necessarily correspond to the strongest emission intensity. Rather, it reflects whether the spectral feature at that position is sufficiently resolved, less affected by neighboring interference, and able to track concentration variation stably.

Considering that Grad-CAM mainly uses gradient information to localize the input regions to which the convolutional feature extractor is sensitive, we further calculated a wavelength-wise contribution profile from the Transformer output and the Yb regression head. Let the output sequence of the Transformer be

$$\mathbf{Z} \in \mathbb{R}^{B \times L' \times D} \quad (12)$$

where B is the batch size, L' is the compressed spectral sequence length, and D is the Transformer embedding dimension. The sequence was flattened before being passed to the Yb specific linear regression head. Accordingly, the regression head weight vector was reshaped as

$$\mathbf{W}^{\text{Yb}} \in \mathbb{R}^{L' \times D} \quad (13)$$

For sample b , the contribution score at spectral position j was calculated as

$$s_{b,j} = \sum_{d=1}^D Z_{b,j,d} W_{j,d}^{\text{Yb}} \quad (14)$$

The nonnegative positional contribution was then obtained as

$$p_{b,j} = \frac{\max(s_{b,j}, 0)}{\sum_{k=1}^{L'} \max(s_{b,k}, 0)} \quad (15)$$

Thus, the positional contributions of each sample summed to one along the compressed spectral axis. The contributions were averaged over the replicate spectra to obtain the mean profile:

$$\bar{p}_j = \frac{1}{B} \sum_{b=1}^B p_{b,j} \quad (16)$$

The resulting profile was interpolated to the input wavelength axis and normalized again after interpolation so that the contributions summed to one. This wavelength-wise profile was used to compare the relative spectral evidence associated with Yb prediction under different matrix conditions.

To examine whether the overall dispersion of the contribution profile was associated with prediction error, the normalized Shannon entropy was calculated for each Yb contribution profile. The normalized entropy was defined as $H_{\text{norm}} = -\sum_{i=1}^N p_i \ln(p_i) / \ln(N)$, where p_i is the normalized contribution at wavelength position i , and N is the number of wavelength sampling points. The normalized entropy values were 0.9494, 0.9637, 0.9628, and 0.9704 for matrices 1-4, respectively. The corresponding Yb RMSE values were 0.421, 0.451, 0.676, and 1.521 wt.%. Matrix 4 showed both the highest entropy and the highest prediction error. The overall dispersion of the contribution profile alone therefore cannot explain the differences in prediction error among the four matrices. **Fig. S6** compares the average Yb contribution profiles across the four matrices. Matrix 2 shows broader high contribution regions around several Yb related bands, whereas Matrix 4 shows a more dispersed profile. Matrix 1 retains clearer responses in the short wavelength region, and Matrix 3 shows more localized maxima. The profiles show that the model used different spectral evidence under different matrix conditions. These differences reflect the influence of matrix composition on the availability and relative usefulness of Yb related spectral features.

Table 3 Atomic parameters of selected Yb lines used for line assignment and attribution discussion.

Wavelength(nm)	Line	$A_{ki}/10^8\text{S}^{-1}$	g_k	Transition Energy (eV)	Transitions
328.937	Yb II	1.62	4	3.768	$4f^{14}6s^2S_{1/2} \rightarrow 4f^{14}6p^2P_{3/2}^0$
369.419	Yb II	1.23	2	3.356	$4f^{14}6s^2S_{1/2} \rightarrow 4f^{14}6p^2P_{1/2}^0$
398.799	Yb I	1.92	3	3.109	$4f^{14}6s^2S_0 \rightarrow 4f^{14}6s6p^1P_1^0$
457.621	Yb I	/	5	2.709	$4f^{14}6s6p^3P_1^0 \rightarrow 4f^{14}6s6d^3D_2$
493.55	Yb I	/	7	2.512	$4f^{14}6s6p^3P_2^0 \rightarrow 4f^{14}6s6d^3D_3$
555.647	Yb I	0.0115	3	2.231	$4f^{14}6s^2^1S_0 \rightarrow 4f^{14}6s6p^3P_1^0$

To relate these profile-level differences to specific transitions, representative wavelength windows were then examined in **Fig. 5** with reference to **Table 3**. These windows cover Yb II 328.937 and 369.419 nm together with Yb I 398.799, 457.621, 493.550, and 555.647 nm. Among the selected lines, 328.937 nm was one of the most stable features. This wavelength corresponds to a strong Yb II transition ($4f^{14}6s^2S_{1/2} \rightarrow 4f^{14}6p^2P_{3/2}^0$) with $A_{ki} = 1.62 \times 10^8 \text{ s}^{-1}$. Because it is a strong allowed ionic transition with the ground state as the lower level, it is more likely to remain observable as long as a sufficient fraction of Yb^+ is present in the plasma. In LIBS, such intrinsically strong ionic lines are generally less affected by moderate background elevation and local spectral crowding and therefore tend to retain better discriminative value across matrices. The consistently high contribution at 328.937 nm is thus consistent with its spectroscopic characteristics. By comparison, 369.419 nm also belongs to Yb II, but

its contribution was weaker and less stable, in line with its lower transition probability and statistical weight.

The Yb I lines showed stronger matrix dependence. The line at 398.799 nm, which has a relatively large transition probability, still contributed in several matrices, but its usefulness appeared to depend more strongly on the balance between neutral and ionic species. The lines at 457.621 nm and 493.55 nm were even more matrix sensitive, with 493.55 nm showing a strong contribution in Matrix-3 but a much weaker response in Matrix-1. This indicates that the effective feature combination used by the model was not identical across matrices. The line at 555.647 nm, with a much smaller A_{ki} value ($0.0115 \times 10^8 \text{ s}^{-1}$), showed the strongest variability, consistent with a weaker intrinsic transition that is more vulnerable to background fluctuation and neighboring interference. Overall, the attribution results indicate that the model assigned higher contributions to several Yb-related spectral regions with clear spectroscopic relevance, although the relative usefulness of individual lines varied with matrix condition. These results indicate that the model used several informative Yb related spectral regions rather than relying on a single universally dominant line, while the relative contribution of individual regions varied with matrix condition.

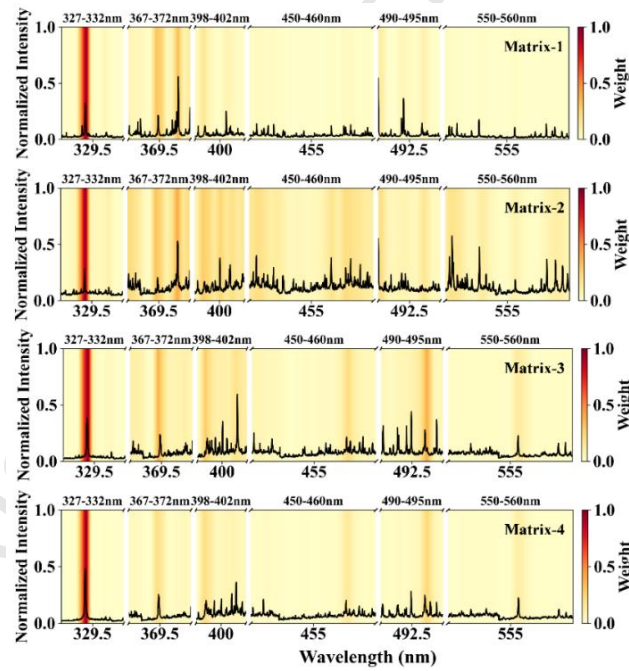


Fig. 5. LIBS spectra and the corresponding wavelength-wise weight distributions under four matrices (Matrix-1 to Matrix-4). The black curves denote normalized spectral intensity, while the heatmap shows the relative weights assigned to each wavelength. Selected wavelength windows (327-332 nm, 367-372 nm, 398-402 nm, 450-460 nm, 490-495 nm, and 550-560 nm) to compare the changes in the model learning of corresponding feature wavelength weights under different matrices.

To further contextualize the Grad-CAM and wavelength-contribution results, a spectral zone SHAP analysis was performed as an independent perturbation-based explanation baseline. Point wise SHAP was not used because adjacent wavelength points in LIBS spectra are highly correlated and single-point perturbations may generate artificial spectral patterns. Instead, the spectrum was divided

into continuous wavelength zones, and the contribution of each zone to the Yb prediction was estimated by spectral-zone perturbation. The spectral-zone SHAP results are summarized in **Table S7** and visualized in Fig. S7. The highest-ranked SHAP zone was 325–330 nm in all four matrices, corresponding to the region containing Yb II 328.937 nm. This result is consistent with the Grad-CAM and wavelength-contribution analysis, in which the 328.937 nm region was also identified as an important Yb related feature. The selected Yb related zones accounted for 0.198-0.277 of the normalized spectral zone SHAP contribution, indicating that a notable fraction of the perturbation-based attribution was concentrated in spectroscopically meaningful Yb regions.

4. Conclusions

This study shows that matrix variation destabilizes LIBS quantification of REEs in ores by altering both the spectral background and the local emission intensity, thereby limiting the transferability of line-based calibration across matrices. The CNN-Transformer fusion model achieved the lowest RMSE in most single-matrix tasks and generally showed more consistent prediction errors than the compared models. For Yb, the only analyte shared by all four matrices, the fusion model also gave the best prediction in the mixed-matrix evaluation. Feature space analysis showed that CNN-Transformer reduced matrix dominated separation while retaining the spectral information required for accurate Yb prediction across the four matrices. The attribution results suggest that this advantage did not arise from reliance on a single dominant line, but from the use of multiple Yb-related spectral features with different degrees of matrix stability. Among them, the persistent contribution near 328.937 nm is consistent with the more favorable spectroscopic properties of this transition. These findings indicate that the fusion model retained concentration relevant spectral information while reducing matrix dominated separation in the learned representation. The fusion model provides a more robust route for LIBS quantification of REEs in polymetallic ores under matrix variation.

CRedit authorship contribution statement

Huihui Zhu: Conceptualization, Investigation, Data analysis, Writing-original draft. **Xuemei Jin:** Review & editing. **Jiawei Wang:** Validation, Visualization. **Yubo Zhang:** Supervision. **Xiaohui Su:** Supervision. **Jian Wu:** Methodology. **Xinyu Guo:** Methodology. **Xuelin Dong:** Resources. **Fuli Chen:** Supervision. **Tao Lü:** Funding acquisition, Writing – review & editing, Supervision. All authors read and approved the final manuscript.

Declaration of competing interest

The authors declare that they have no conflict of interest.

Data availability

Data will be made available on request.

ACKNOWLEDGMENTS

This work was supported by National Key Research and Development Program of China (2024YFF0728300); Key Research and Development Program of Hubei Province (2023BAB063); National Natural Science Foundation of China (42441830); Open Project Program of National Key Laboratory of Uranium Resources Exploration-Mining and Nuclear Remote Sensing (No. NKLUR-2025-YB-007).

REFERENCES

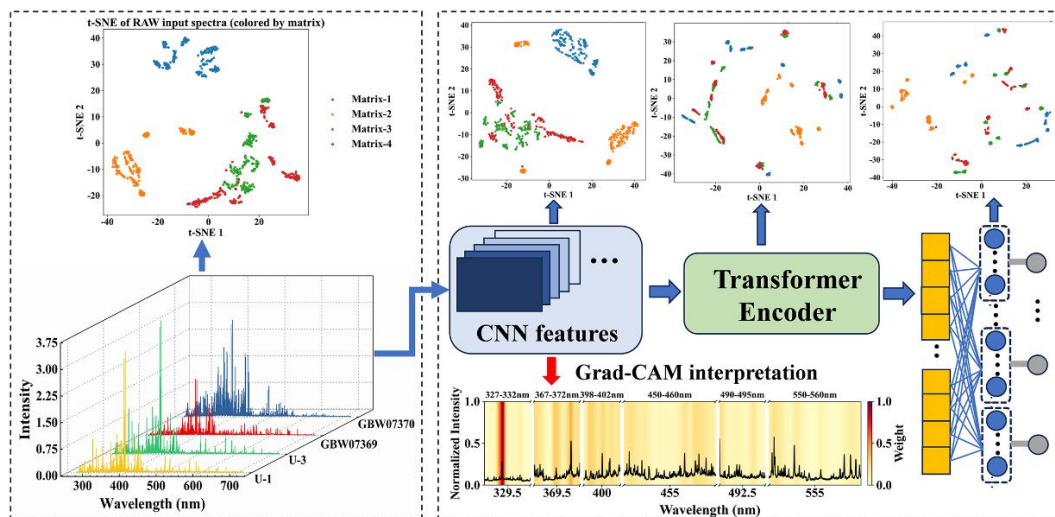
- [1] S. Wang, J.B. Sun, X. Chi, M.J. Zhou, H.Y. Ma, Y. Zhang, J. Liu, High-performance multiphase Sm-Co-B alloys with coercivities up to 6.71 MA·m⁻¹, **Nat. Commun.** 15 (2024) 10158.
- [2] S. Liu, Sm-Co high-temperature permanent magnet materials, **Chin. Phys. B** 28 (2019) 017501.
- [3] G.T. Bholane, T.S. Ganesapandy, A.B. Phatangare, F.M.D. Attar, S.S. Dahiwalé, S.V. Suryanarayana, V.N. Bhoraskar, S.D. Dhole, Activation cross section measurements and estimation of photon and neutron induced nuclear reactions for ytterbium isotopes with covariance analysis, **Radiat. Phys. Chem.** 195 (2022) 110066.
- [4] T. Zhong, J.M. Kindem, E. Miyazono, A. Faraon, Nanophotonic coherent light-matter interfaces based on rare-earth-doped crystals, **Nat. Commun.** 6 (2015) 8206.
- [5] E. Kendir Tekgül, B. Ortaç, Rapid photo-bleaching gamma irradiated Yb-doped optical fibers by high-energy ns pulsed laser, **Opt. Fiber Technol.** 81 (2023) 103565.
- [6] V. Balaram, Rare earth elements: A review of applications, occurrence, exploration, analysis, recycling, and environmental impact, **Geosci. Front.** 10 (2019) 1285–1303.
- [7] P. Chen, E.S. Ilton, Z. Wang, K.M. Rosso, X. Zhang, Global rare earth element resources: A concise review, **Appl. Geochem.** 175 (2024) 106158.
- [8] V. Palleschi, S. Legnaioli, F. Poggialini, F.O. Bredice, I.A. Urbina, N. Lellouche, S. Messaoud Aberkane, Laser-induced breakdown spectroscopy, **Nat. Rev. Methods Primers** 5 (2025) 17.
- [9] Z. Wu, J. Wu, X. Guo, H. Zhu, Y. Zhang, X. Su, F. Chen, M. Li, R. Wang, K. Xu, T. Lü, Laser-induced breakdown spectroscopy combined with multi-task convolutional neural network for analyzing Sm, Nd, and Gd elements in uranium polymetallic ore, **Spectrochim. Acta Part B At. Spectrosc.** 226 (2025) 107153.
- [10] D.W. Hahn, N. Omenetto, Laser-induced breakdown spectroscopy (LIBS), Part II: Review of instrumental and methodological approaches to material analysis and applications to different fields, **Appl. Spectrosc.** 66 (2012) 347–419.
- [11] R.S. Harmon, G.S. Senesi, Laser-induced breakdown spectroscopy – A geochemical tool for the 21st century, **Appl. Geochem.** 128 (2021) 104929.
- [12] T.F. Akhmetzhanov, A.M. Popov, Direct determination of lanthanides by LIBS in REE-rich ores: comparison between univariate and DoE based multivariate calibrations with respect to spectral resolution, **J. Anal. At. Spectrom.** 37 (2022) 2330–2339.
- [13] T.A. Labutin, S.M. Zaytsev, A.M. Popov, N.B. Zorov, A novel approach to sensitivity evaluation of laser-induced breakdown spectroscopy for rare earth elements determination, **J. Anal. At. Spectrom.** 31 (2016) 2223–2226.

- [14] D.S. Ferreira, D.V. Babos, M.H. Lima-Filho, H.F. Castello, A.C. Olivieri, F.M. Verbi Pereira, E.R. Pereira-Filho, Laser-induced breakdown spectroscopy (LIBS): calibration challenges, combination with other techniques, and spectral analysis using data science, **J. Anal. At. Spectrom.** 39 (2024) 2949–2973.
- [15] E. Képeš, I. Gornushkin, P. Pořízka, J. Kaiser, Spatiotemporal spectroscopic characterization of plasmas induced by non-orthogonal laser ablation, **Analyst** 146 (2021) 920–929.
- [16] D.S. Vogt, S. Schröder, S. Frohmann, P.B. Hansen, F. Seel, M. Gensch, H.-W. Hübers, Spatiotemporal characterization of the laser-induced plasma plume in simulated Martian conditions, **Spectrochim. Acta Part B At. Spectrosc.** 187 (2022) 106326.
- [17] C.R. Bhatt, F.Y. Yueh, J.P. Singh, Univariate and multivariate analyses of rare earth elements by laser-induced breakdown spectroscopy, **Appl. Opt.** 56 (2017) 2280–2287.
- [18] N. Rethfeldt, P. Brinkmann, D. Riebe, T. Beitz, N. Köllner, U. Altenberger, H.G. Löhmannsröben, Detection of rare earth elements in minerals and soils by laser-induced breakdown spectroscopy (LIBS) using interval PLS, **Minerals** 11 (2021) 1379.
- [19] S. Ahlawat, A. Singh, S. Sahu, P.K. Mukhopadhyay, R. Arya, S.K. Dixit, Trigger-free LIBS using kHz and a few mJ laser in combination with random forest regression for the quantitative analysis of steel elements, **J. Anal. At. Spectrom.** 39 (2024) 1927–1937.
- [20] P. Dehbozorgi, L. Duponchel, V. Motto-Ros, T.W. Bocklitz, Enhancing prediction stability and performance in LIBS analysis using custom CNN architectures, **Talanta** 284 (2025) 127192.
- [21] X. Fang, H. Yu, Q. Huang, Z. Jiang, D. Pan, W. Gui, Full-spectrum LIBS quantitative analysis based on heterogeneous ensemble learning model, **Chemom. Intell. Lab. Syst.** 257 (2025) 105321.
- [22] J. Yang, S. Li, Z. Zhang, D. Qian, S. Sun, X. Liu, Z. Liu, Transformer-based deep learning models for quantification of La, Ce, and Nd in rare earth ores using laser-induced breakdown spectroscopy, **Talanta** 292 (2025) 127937.
- [23] C. Zhang, W. Song, Y. Lyu, Z. Liu, X. Gao, Z. Hou, Z. Wang, Dual-branch convolutional neural network with attention modules for LIBS-NIRS data fusion in cement composition quantification, **Anal. Chim. Acta** 1351 (2025) 343899.
- [24] X. Su, F. Chen, H. Zhu, Y. Zhang, X. Zhang, B. Tang, J. Wu, X. Guo, J. Wang, X. Jin, T. Lü, Laser-induced breakdown spectroscopy combined with multi-scale convolutional neural network and attention mechanisms for analyzing Dy, Nd, and Yb in uranium polymetallic ores, **Anal. Chim. Acta** 1386 (2026) 345035.
- [25] P. Yaroshchyyk, J.E. Eberhardt, Automatic correction of continuum background in laser-induced breakdown spectroscopy using a model-free algorithm, **Spectrochim. Acta Part B At. Spectrosc.** 99 (2014) 138–149.
- [26] K. Liu, R. Zhou, W. Zhang, Z. Tang, J. Yan, M. Lv, X. Li, Y. Lu, X. Zeng, Interference correction for laser-induced breakdown spectroscopy using a deconvolution algorithm, **J. Anal. At. Spectrom.** 35 (2020) 762–766.
- [27] W. Zhao, C. Li, C. Yan, H. Min, Y. An, S. Liu, Interpretable deep learning-assisted laser-induced breakdown spectroscopy for brand classification of iron ores, **Anal. Chim. Acta** 1166 (2021) 338574.
- [28] X. Liu, H. An, W. Cai, X. Shao, Deep learning in spectral analysis: Modeling and imaging, **TrAC Trends Anal. Chem.** 172 (2024) 117612.
- [29] E. Képeš, J. Vrábek, T. Brázdil, P. Holub, P. Pořízka, J. Kaiser, Interpreting convolutional neural network classifiers applied to laser-induced breakdown optical emission spectra, **Talanta** 266 (2024) 124946.

- [30] G.Y. Shi, H.P. Wu, S.H. Luo, X.Y. Lu, B. Ren, Q. Zhang, W.Q. Lin, R.Y. Chen, P. Guo, H. Bin Chen, Z.Q. Tian, G.F. Shao, L. Yang, G.K. Liu, 1D Gradient-Weighted Class Activation Mapping, Visualizing Decision Process of Convolutional Neural Network-Based Models in Spectroscopy Analysis, **Anal. Chem.** 95 (2023) 9959–9966.
- [31] Y. LeCun, Y. Bengio, G. Hinton, Deep learning, **Nature** 521 (2015) 436–444.
- [32] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, Ł. Kaiser, I. Polosukhin, Attention is All You Need, in: I. Guyon, U. Von Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, R. Garnett (Eds.), **Adv. Neural Inf. Process. Syst.** 30 (2017) 5998–6008.
- [33] L. van der Maaten, G. Hinton, Visualizing data using t-SNE, **J. Mach. Learn. Res.** 9 (2008) 2579–2605.

Graphical abstract

Full-spectrum LIBS spectra from four polymetallic ore matrices are processed by a CNN feature extractor and a Transformer encoder for robust concentration prediction, while Grad-CAM attribution highlights key wavelength regions consistent with characteristic emission features and t-SNE visualizations indicate reduced matrix-driven clustering.



Highlights

- Developed a fusion model for LIBS of rare earth elements under matrix variation.
- The fusion model outperforms single-branch deep learning models.
- CNN-Transformer improves robustness against matrix variation.
- Interpretability analysis relates robustness to stable analyte-related features.